Laboratory Exercise
The Model Matrix

Objective:

At the end of the exercise, the students should be able to:

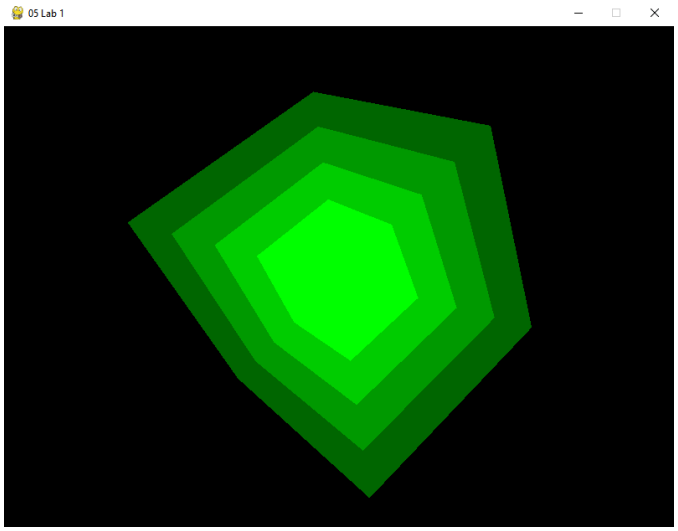
- Replicate 3D objects using a matrix stack.

Software Requirement:

- Python 3.7 or higher

Procedure:

1. Launch **IDLE** and open your Python module from **04 Laboratory Exercise 1**.
2. Save a copy of your module using a different filename, then run it by clicking **Run** > **Run Module** or pressing **F5**.
3. Change the window's caption to **`"05 Lab 1"`**.
4. Remove all the **`glColor3f()`** statements from your **`draw_cube()`** function.
5. Add a new function named **`draw_object()`**. This function will allow you to create four (4) cubes of different sizes and colors.
6. In your **`draw_object()`** function, create four (4) pairs of **`glPushMatrix()`** and **`glPopMatrix()`** statements. **`glPushMatrix()`** pushes the current matrix stack down by one, duplicating the current matrix. After a **`glPushMatrix()`** call, the matrix on top of the stack is identical to the one below it. **`glPopMatrix()`** pops the current matrix stack, replacing the current matrix with the one below it on the stack.
7. In between your **`glPushMatrix()`** and **`glPopMatrix()`** statements, perform the following:
    - Change the size of the cube.
    - Color the cube.
    - Call your **`draw_cube()`** function.



8. In your while loop, call the **`draw_object()`** function instead of **`draw_cube()`**.
9. Show/submit your code and output to your instructor.
10. Save a copy of your code for future use.

Grading Rubric:

CRITERIA	PERFORMANCE INDICATORS	POINTS
Correctness	The geometric transformations applied to the cube are complete and correct.	30
Efficiency	The code is concise without sacrificing correctness and logic.	20
TOTAL		**50**